

# Aryaman Gokarn

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## EDUCATION

### University of California, Los Angeles (UCLA)

Expected: Dec 2024

*Master of Science in Electrical and Computer Engineering (Research: Diffusion Models)*

*CGPA: 3.95/4.0*

*Courses: Neural Network and Deep Learning, Large Scale Data-Mining, Computational Imaging*

### Sardar Patel Institute of Technology

May 2023

*Bachelor of Technology in Electronics & Telecommunication with Minors in Computer Science*

*CGPA: 9.45/10.0*

*Courses: Data Structures & Algorithms, Web Technologies, System Design, AI & ML*

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, R, SQL, Java, JavaScript, MATLAB, HTML, CSS

**Libraries & Frameworks:** TensorFlow, Pytorch, JAX, Scikit-learn, CUDA, OpenCL, ONNX, React, Node.js

**Developer Tools:** AWS, Azure, Google Cloud Platform, Power BI, MongoDB, Git, Docker, Kubernetes, LLMs

**Other:** Generative AI, MLOps, Deep Learning, Computer Vision, Data Science, RLHF, Distributed Systems, Research

## WORK EXPERIENCE

### Hyphenova

Jun 2024 – Sept 2024

*Software Development Engineer Intern*

*Los Angeles, USA*

- Achieved **25%** reduction in page load times by optimizing **React** code and **Node.js** for app landing pages.
- Streamlined **data processing** by **30%** by optimizing **deployment pipelines**, attaining **15%** less data redundancy
- Performed **code reviews**, ensured **code hygiene**, conducted **bug triages**, and provided feedback on app features

### Velocity Global Solutions

Mar 2023 – May 2023

*Machine Learning Engineer Intern*

*Mumbai, India*

- Developed a **multimodal pipeline** (text, images, audio) to identify human sentiment, using **NLP** and **CNN**.
- Designed and implemented end-to-end object detection system using **YOLOv8** and **Faster R-CNN**, resulting in **12% increase** in accuracy and deployed the UI using Streamlit.
- Improved image colorization by **15%** using **VAE & GAN**, enhancing automated image enhancement app.
- Collaborated with **20+ clients** to assess their business needs and boosted operational efficiency by **15%**.

### Deloitte

Jan 2022 – Jun 2022

*IT and Specialized Assurance Intern*

*Mumbai, India*

- Enhanced **security** by evaluating **71 applications** and **136 controls** of **India's largest** private sector bank.
- Reduced **system downtime** by **25%** through **IT infrastructure assessments**, saving clients **\$20,000** annually

### Maven AI Technologies

Jun 2021 – Sept 2021

*Machine Learning Intern*

*Mumbai, India*

- Automated **extraction** of data from **10+** websites, resulting in a **20%** reduction in **data collection** time.
- Enhanced deep learning model's accuracy by **15%** through accurate annotation of **10,000+** data points per week.
- Collaborated with data scientists, **finetuning 5+** deep learning models that attained **97%** precision and recall.
- Produced data **visualizations** and **summaries** of data characteristics, increasing communication efficiency by **8%**

## RESEARCH PROJECTS

### Semantic Segmentation for Advanced Land Cover Mapping | Published in IEEE 5th ICIRCA

- Developed a novel semantic segmentation approach, by leveraging base models (**VGG16** and **Resnet50**) as the backbone of the segmentation model (**U-Net**) and achieving an overall classification accuracy of **96.63%**.

### Covid-19 Radiography using ConvNets | Published in IEEE 4th SUMMA

- Engineered an AI-driven diagnostic tool using ConvNets that analyzed over **5,000+** chest radiographs, achieving a **95.45%** accuracy rate in detecting COVID-19 abnormalities, significantly aiding rapid response initiatives.

### Real-Time Counterfeit Currency Detection System | Python, OpenCV, Pytesseract

- Improved currency security by implementing an image processing pipeline to authenticate Indian notes, scrutinizing security features. Attained a remarkable **95%** accuracy rate, ensuring robust detection of counterfeit bills.

## LEADERSHIP AND INVOLVEMENT

- Guided **60+ underrepresented women and minorities in Tech** by teaching **Foundations of ML**.
- Led **25** team members to **build** nationwide community of **entrepreneurship-driven** students as GS of 100xVC.
- Achieved top **5%** rank across India in e-Yantra Robotics Competition by building an autonomous bot for industry.